

PIZZA



Sales Analysis





Pizza Sales Analysis (SQL Project) – Introduction

This Pizza Sales Analysis project is performed using SQL to explore and analyze pizza store sales data. The main goal of this project is to extract meaningful insights such as **total revenue, total orders, top-selling pizzas, category-wise sales performance**, and **peak ordering time**.

By writing SQL queries with JOIN, GROUP BY, ORDER BY, Aggregate functions, and Window functions, this analysis helps to understand customer buying patterns and supports business decisions like improving menu strategy, increasing revenue, and managing inventory efficiently.

1. Retrieve the total number of orders placed

```
select count(order_id) as Total_Orders from
orders;
```

	total_orders bigint
1	21350



2. Calculate the total revenue generated from the pizza sales

```
select round(sum(pizza.price*order_detail.quantity)::numeric,2)
as Total_Revenue from order_detail
join pizza on
pizza.pizza_id=order_detail.pizza_id;
```

	total_revenue numeric
1	817860.05



3. Identify the highest-price pizza

```
select pizza_type.name as Pizza,pizza.price as Highest_price
      from pizza_type
      join pizza on
      pizza.pizza_type_id=pizza_type.pizza_type_id
 group by 1,2 order by Highest_price desc limit 1;
```



	pizza character varying 	highest_price double precision 
1	The Greek Pizza	35.95

4. Identify the most common pizza size order

```
select pizza.size,count(order_detail.order_details_id) as
      Total_Order from pizza
      join order_detail on
      order_detail.pizza_id=pizza.pizza_id
group by 1 order by Total_Order desc;
```

	size character varying 	total_order bigint 
1	L	18526
2	M	15385
3	S	14137
4	XL	544
5	XXL	28



5 list the top most ordered pizza type along with their quantity

```
select pizza_type.name,sum(order_detail.quantity) as Total_Order
      from pizza_type
      join pizza on
pizza_type.pizza_type_id=pizza.pizza_type_id
      join order_detail on
      order_detail.pizza_id=pizza.pizza_id
group by 1 order by Total_Order desc limit 5;
```

	name character varying	total_order bigint
1	The Classic Deluxe Pizza	2453
2	The Barbecue Chicken Pizza	2432
3	The Hawaiian Pizza	2422
4	The Pepperoni Pizza	2418
5	The Thai Chicken Pizza	2371



6.Find the Total quantity of each pizza category ordered

```
select pizza_type.category as
Pizza_Category,sum(order_detail.quantity) as Total_Order from
    pizza_type
    join pizza on
pizza.pizza_type_id=pizza_type.pizza_type_id
    join order_detail on
    order_detail.pizza_id=pizza.pizza_id
group by 1 order by Total_Order desc;
```



	pizza_category character varying 	total_order bigint 
1	Classic	14888
2	Supreme	11987
3	Veggie	11649
4	Chicken	11050

7. Determine the distribution of orders by hour of the day

```
select extract(hour from time) as Hour,count(orders.order_id) as  
       Total_Order from orders  
group by 1 order by Total_Order desc limit 5;
```

	hour numeric 	total_order bigint 
1	12	2520
2	13	2455
3	18	2399
4	17	2336
5	19	2009



8.Find the category wise distribution

```
select pizza_type.category,count(name) from pizza_type
group by 1;
```

	category character varying 	count bigint 
1	Supreme	9
2	Chicken	6
3	Classic	8
4	Veggie	9



9.Group the order by date and calculate the avg number of pizza ordered per day

```
select round(avg(Quantity),2) as Avg_Orders from(  
select orders.date,sum(order_detail.quantity) as Quantity from  
orders  
join order_detail on  
order_detail.order_id=orders.order_id  
group by 1);
```



	avg_orders numeric
1	138.47

10. Determine the top 3 most ordered pizza type based on revenue

```
select
pizza_type.name,sum(pizza.price*order_detail.quantity) as
Revenue from pizza_type
join pizza on
pizza_type.pizza_type_id=pizza.pizza_type_id
join order_detail on
pizza.pizza_id=order_detail.pizza_id
group by 1 order by Revenue desc limit 3;
```

	name character varying 	revenue double precision 
1	The Thai Chicken Pizza	43434.25
2	The Barbecue Chicken Pizza	42768
3	The California Chicken Pizza	41409.5



11.Calculate the % contribution of each pizza type of total revenue

```
select
pizza_type.category,round((sum(pizza.price*order_detail.quantity)/(select
sum(pizza.price*order_detail.quantity) from order_detail
join pizza on pizza.pizza_id=order_detail.pizza_id)*100)::numeric,2) as
Revenue from pizza_type
join pizza on
pizza.pizza_type_id=pizza_type.pizza_type_id
join order_detail on
pizza.pizza_id=order_detail.pizza_id
group by 1 order by Revenue desc;
```



	category character varying 	percentage numeric 
1	Classic	26.91
2	Supreme	25.46
3	Chicken	23.96
4	Veggie	23.68

12. Analysis the cumulative revenue generated over time

```
select Sales.date,round(sum(Sales.Revenue) over(order by
Sales.date)::numeric,2) as cumulative_Revenue from(
select orders.date,sum(order_detail.quantity*pizza.price) as
Revenue from order_detail
join pizza on
pizza.pizza_id=order_detail.pizza_id
join orders on
orders.order_id=order_detail.order_id
group by 1) as Sales;
```



	date date	cumulative_revenue numeric
1	2015-01-01	2713.85
2	2015-01-02	5445.75
3	2015-01-03	8108.15
4	2015-01-04	9863.60
5	2015-01-05	11929.55

13.Determine the top 3 most orderd pizza type based on revenue
foe each pizza category

```
select name,category,Rank from
(select category,name,Revenue,Rank() over(partition by category
order by Revenue desc) as Rank from(
select
pizza_type.category,pizza_type.name,sum(pizza.price*order_detail.qua
ntity) as Revenue from pizza_type
join pizza on
pizza.pizza_type_id=pizza_type.pizza_type_id
join order_detail on
order_detail.pizza_id=pizza.pizza_id
group by 1,2) as b) as a where Rank<=3;
```



	name character varying 	category character varying 	rank bigint 
1	The Thai Chicken Pizza	Chicken	1
2	The Barbecue Chicken Pizza	Chicken	2
3	The California Chicken Pizza	Chicken	3
4	The Classic Deluxe Pizza	Classic	1
5	The Hawaiian Pizza	Classic	2
6	The Pepperoni Pizza	Classic	3



THANK YOU